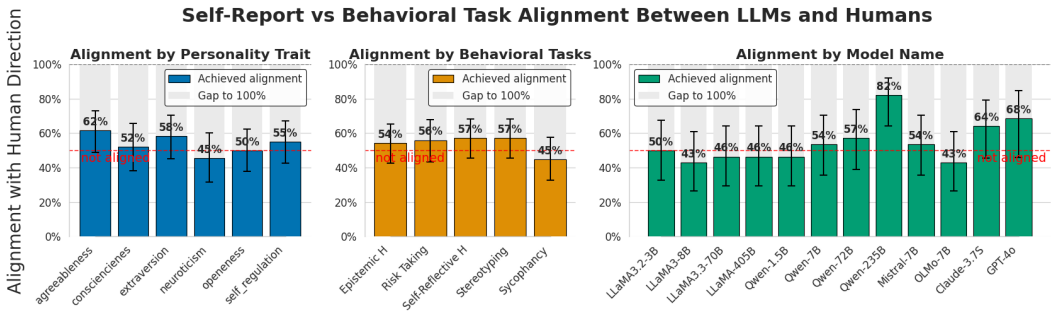


Figure 3: **Alignment Between LLMs and Humans Across Personality Traits, Behavioral Tasks, and Model Types.** Each panel shows the percentage of cases where LLM self-reports were directionally aligned with behavioral task in accordance with directions expected from human subjects (*Achieved alignment*, colored bars), with the remaining proportion indicating the *Gap to 100%* (light shading). The first panel summarizes alignment in expected association between self-reports and behavioral tasks by self-reported **personality traits**, the second by **behavioral task**, and the third by **model name**, grouped by model family and ordered by increasing parameter size. Percentages above bars indicate the exact alignment proportion. Line at 50% represents random behavior (i.e., % alignment expected by chance). Error bars represent 95% confidence intervals (CIs).

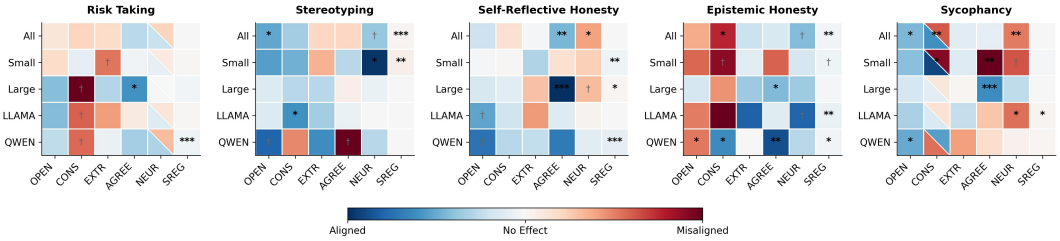


Figure 4: **Alignment based on Mixed-Effects Models estimating LLM Personality Trait Effects on Task Behavior.** Each panel shows mixed-effects model coefficients for LLMs' self-reported personality traits predicting behavior across five tasks, with results presented for all models, small models, large models, the LLaMA family, and the Qwen family. Blue cells indicate effects **aligned** with human expectations, while red cells indicate effects in the opposite direction. Split diagonal cells mark cases where human expectations are unclear; blue is on top for positive coefficients and on the bottom for negative. **Color intensity** reflects effect magnitude, with darker shades indicating stronger effects. **Significance** is denoted as † $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$. The detailed numerical values are provided in Table 3 in the Appendix C.

are also grouped by *Small* and *Large* models and by *Qwen* and *LLaMA* families for which we have 4 individual LLMs of varying sizes. We observe local, non-systematic patterns of partial alignment between self-reported *Openness* and behavioral tasks around *Stereotyping*, *Self-Reflective Honesty*, and *Sycophancy* (uniformly blue columns), though effects rarely reach statistical significance. For *Epistemic Honesty* we observe alignment with self-reported *Extroversion*, *Neuroticism*, and *Self-regulation* (uniformly blue columns), but again with few statistically significant associations. At the LLM-family level, *Qwen family* uniquely displays consistent alignment of all self-reported traits with *Self-Reflective Honesty*. Still, these results underscore that ***alignment patterns are rare and inconsistent***, with both alignment and misalignment varying across traits, tasks, and architectures.

These results highlight that ***LLMs' self-reported traits rarely translate into behavior–alignment hovers near chance for small–mid models and is sporadic even for frontier ones*** (with only a narrow, isolated exception). This dissociation between linguistic self-presentation and action limits behavioral controllability and weakens questionnaires as proxies for downstream behavior.

## 4 RQ3: CONTROLLABILITY

RQ2 revealed that LLMs exhibit stable and coherent self-reported personality traits, but these do not reliably predict behavior in downstream tasks. When associations are statistically significant, they frequently diverge from patterns observed in human behavioral psychology. This suggests a fundamental disjunction: unlike humans, LLMs lack intrinsic goals, motivations, or consistent internal states, and their behavior appears more contingent on prompt structure and context than on stable traits. ***Instructional alignment may shape self-reports, but this alignment is often superficial.*** For example, a model that self-reports low risk-taking may still act inconsistently in decision-making contexts. Such inconsistencies highlight the fragility of LLM personality expressions and suggest that self-reports alone are poor indicators of behavioral tendencies. Given this, we ask: if self-reports are unreliable, can we instead control behavior more directly? Specifically, can targeted interventions—such as persona injection—shape both trait self-reports and real-world task behaviors in more human-like and consistent ways?

**Research Question 3** (Control)**.** *How do intervention methods (e.g., persona injection) influence self-reported trait profiles and their behavioral manifestations?*

### 4.1 EXPERIMENT SETUP

To evaluate our research question, we replicate RQ1 and RQ2 procedures, using the BFI and SRQ questionnaires for self-reports and two behavioral tasks—sycophancy and risk-taking—that showed the most counterintuitive patterns in RQ2. While self-regulation is typically linked to reduced risk-taking in humans (Duell et al., 2016), and agreeableness predicts sycophantic tendencies (Nettle & Liddle, 2008), these associations were weak or absent in RQ2.

Instead of default personas, we introduce *trait-specific personas* to test whether explicit personality prompting enhances alignment between self-reports and behavior. We conduct two experiments: **(1) Agreeableness Persona**, assessing its impact on self-reported traits and sycophantic behavior; and **(2) Self-Regulation Persona**, evaluating effects on self-reports and risk-taking behavior. Personas are constructed by sampling representative trait keywords, following **three different prompting strategies** established in prior LLM personality research (Jiang et al., 2024a; Serapio-García et al., 2023; Dash et al., 2025). Implementation details are provided in Table 10 in the Appendix H.

### 4.2 STATISTICAL ANALYSIS

We test whether LLMs exhibit systematic differences in self-reported traits and real-world behaviors before and after trait-specific persona injection. For each of the three prompting strategies, we fit separate binomial logistic regression models to predict persona condition (trait-specific persona vs. default). For the self-report analysis, all six trait scores are used as predictors. For the behavioral analysis, we use the downstream task performance (sycophancy or risk-taking) as a single predictor. All predictors are standardized, and within each prompting strategy, we include prompt variation, sampling temperature, and model as control variables. Inference is based on Wald z-statistics and 95% confidence intervals, shown in Figure 5.

### 4.3 RESULTS

**Self-Report.** ***Trait-specific personas lead to strong alignment on their target traits.*** When injecting the agreeableness persona, logistic regression reveals a significant increase in self-reported agreeableness ($\beta \approx 3.6$ to $4.4$, $p < .001$). Similarly, injecting the self-regulation persona results in a significant increase in self-reported self-regulation ($\beta \approx 2.2$ to $2.9$, $p < .05$). These results confirm that self-reported traits reliably reflect the intended persona in self-report scenarios.

However, ***the inter-trait relationships do not fully align with the patterns observed in RQ1*** (Figure 2), where extraversion, openness, conscientiousness, and agreeableness were meaningfully positively correlated, and neuroticism was negatively associated. In contrast, we find that injecting agreeableness produces an inconsistent effect on self-regulation ($\beta \approx -0.44$ to $0.50$, some n.s., up to $p < .05$), while injecting self-regulation reduces agreeableness ($\beta \approx -1.1$ to $-1.8$, $p < .05$) and openness ($\beta \approx -2.2$ to $-2.8$, $p < .001$). Additionally, the self-regulation persona has little and often non-significant effect on neuroticism or extraversion. Notably, conscientiousness shows a strong and

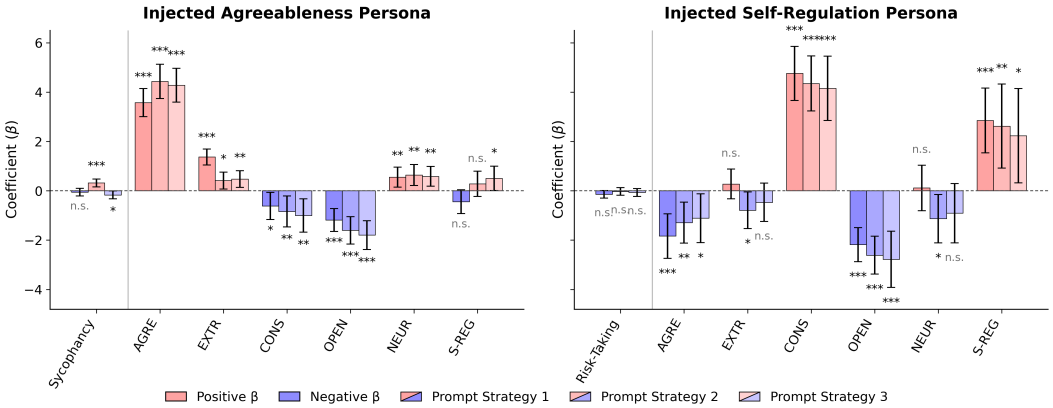


Figure 5: **Trait-Specific Personas Are Detectable via Self-Reports but Not Behavior.** Coefficient estimates (95% CI) from logistic regressions predict persona condition (Agreeableness or Self-Regulation vs. Default) using either six self-reported traits or one behavioral measure (sycophancy or risk-taking). Results are shown across three prompting strategies, indicated by color intensity (Appendix H). Significance levels (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, n.s.) are marked on each bar. Across strategies, self-reports reliably reveal persona presence, whereas behavioral measures do not, indicating limited transfer of persona effects to downstream behavior.

significant increase when the self-regulation persona is applied ($\beta \approx 4.2$ to $4.8$, $p < .001$), exceeding even the effect on self-regulation itself.

**Behavioral Task.** In contrast to the strong alignment observed in self-reports, ***behavioral measures show limited sensitivity to persona injection.*** When using downstream behavior to predict whether a persona was applied, logistic regression models yield mostly non-significant results for both cases. Specifically, sycophantic responses provide weak and inconsistent evidence for predicting whether the agreeableness persona was used ($\beta \approx -0.05$ to $0.32$, n.s. to $p < .001$), and risk-taking behavior similarly fails to reliably distinguish the self-regulation condition ($\beta \approx -0.14$ to $0.20$, n.s.).

These findings suggest that while ***LLMs exhibit clear changes in how they self-report personality traits under different personas, those changes do not consistently manifest in behavior.*** The weak predictive power of real-world tasks highlights a key limitation in the behavioral controllability of LLMs: surface-level trait alignment does not necessarily translate to deeper, goal-driven consistency. This points to a dissociation between linguistic self-presentation and action-oriented decision behavior.

## 5 Discussion

Our study reveals a notable gap between surface-level trait expression and actual behavior in LLMs. Although instruction tuning and persona prompts stabilize self-reported traits, these do not reliably translate to consistent downstream behavior. This challenges the view of LLMs as behaviorally grounded and suggests that current alignment methods favor linguistic plausibility over functional reliability. We discuss this dissociation across three dimensions: *(1)* linguistic–behavioral divergence, *(2)* diagnosis through psychologically grounded frameworks, and *(3)* the illusion of coherence created by current alignment and prompting.

**Linguistic-Behavioral Dissociation in LLMs.** Our findings highlight a dissociation between linguistic self-expression and behavioral consistency in LLMs. While LLMs can simulate personality traits through language (Cao & Kosiński, 2023), these traits likely arise from surface-level pattern matching rather than internalized motivations—unlike human personality, which is grounded in cognitive and affective processes (McCrae & John, 1992). Moreover, LLMs lack temporal consistency and exhibit high prompt sensitivity (Bodroža et al., 2024a). This disconnect is further supported by recent findings that survey-based evaluations—though often linguistically coherent—fail to predict